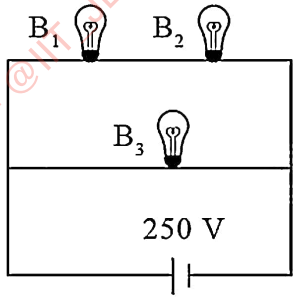
A uniform cubical box of side a is placed on a rough floor and it is to be moved by applying minimum possible force F at a point which is at a distance b above its centre of mass. If the coefficient of friction is $\mu = 0.4$, then what is the maximum possible value of $100 \times \frac{b}{a}$ for which the box will not topple before moving?

- 60
- 50
- 45
- 40

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A 100 W bulb B_1 and two 60 W bulb B_2 and B_3 are connected to a 250 V source as shown in the figure. Now W_1, W_2 and W_3 are the output powers of the bulbs B_1, B_2 and B_3 respectively. Then



- $W_1 > W_2 = W_3$
- $W_1 > W_2 > W_3$
- $W_1 < W_2 = W_3$
- $W_1 < W_2 < W_3$

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The ratio of accelerations due to gravity $g_1 : g_2$ on the surfaces of two planets is $5 : 2$ and the ratio of their respective average densities $\rho_1 : \rho_2$ is $2 : 1$. What is the ratio of respective escape velocities $v_1 : v_2$ from the surface of the planets?

- $5 : 2$
- $\sqrt{5} : \sqrt{2}$
- $5 : 2\sqrt{2}$
- $25 : 4$

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Proton striking a stationary lithium target activate a reaction : $Li^7 + p \rightarrow Be^7 + n$
At what value of proton's kinetic energy can the resulting neutron be stationary.
Given: $m\left({}_1^1H\right) = 1.00783$ amu, $m\left(Be^7\right) = 7.01693$, $m\left(Li^7\right) = 7.01601$, $m(n) = 1.00867, 1$ amu $= 931$ MeV

- 0.2 MeV
- 1.9 MeV
- 3.8 MeV
- 7 MeV

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Maximum kinetic energy of a photoelectron is E, when the wavelength of incident light is λ. If energy becomes 4 times, when wavelength is reduced to $\frac{1}{3}\lambda$, the work-function is....

- $\frac{3hc}{\lambda}$
- $\frac{hc}{\lambda}$
- $\frac{hc}{2\lambda}$
- $\frac{hc}{3\lambda}$

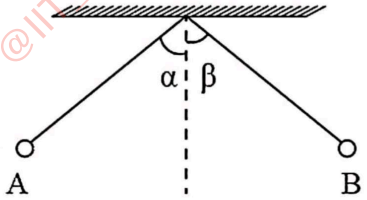
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A mobile phone of mass 100 g falls from the hand of a person through the window of a train. The train is accelerating at $1\ \mathrm{m\ s^{-2}}$ and just 20 s have passed since it left the station from standstill. If the window is 2 m above the ground. With what kinetic energy will the phone will hit the ground? $\left(g = 10\ \mathrm{m\ s^{-2}}\right)$

- ○ 11 J
- ○ 22 J
- ○ 33 J
- ○ 44 J

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Two identical simple pendulums A and B have same point of suspension, having length ℓ each. They are displaced by an angle α and β (α and β are very small and $\beta > \alpha$) and released from rest. Find the time after which B reaches at its initial position for the first time. Assume collision to be elastic and both pendulums move in same plane.



- $\pi\sqrt{\frac{\ell}{g}}$
- $2\pi\sqrt{\frac{\ell}{g}}$
- $\frac{\pi\beta}{\alpha}\sqrt{\frac{\ell}{g}}$
- $\frac{2\pi\beta}{\alpha}\sqrt{\frac{\ell}{g}}$

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The ratio of the speed of sound in a monatomic gas at $27^{\circ}C$ and rms speed of the molecules of the same gas at a temperature of $127^{\circ}C$ is--

- 1 : 2
- $\sqrt{5} : \sqrt{12}$
- 3 : 4
- $\sqrt{13} : \sqrt{17}$

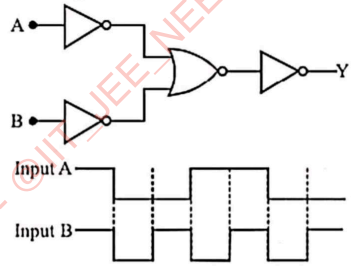
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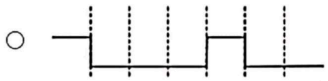
An automatic gun fires 360 bullets per minute with a speed of 360 km/h. If each bullet weighs 20 g, the power of the gun is:

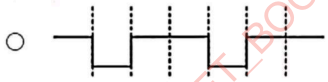
- ○ 75 W
- ○ 150 W
- ○ 300 W
- ○ 600 W

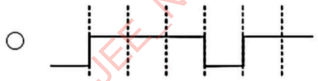
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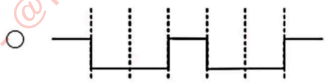
The logic circuit shown has the input waveforms 'A' and 'B' as shown. Pick out the CORRECT output waveform :-











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A bird is flying $3m$ above the surface of water. If the bird is diving vertically down with speed $= 6$ m/s, his apparent velocity as seen by a stationary fish is : $\left(\mu = \frac{4}{3}\right)$

- 12
- 6
- 4
- 8

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Relative density of the material of a body is found by weighing it first in air and then in water. The weight in air is (5.00 ± 0.05) newton and weight in water is (4.00 ± 0.05) newton. Then the relative density of material should be:-

- $5.0 \pm 11\%$
- $5.0 \pm 1\%$
- $5.0 \pm 6\%$
- $1.25 \pm 5\%$

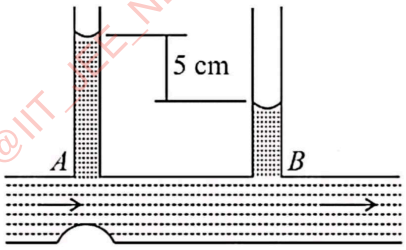
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32 g of O_2 is contained in a cubical container of side 1 m and maintained at a temperature of 127 $^{\circ}$C. The isothermal bulk modulus of elasticity of the gas is (universal gas constant = R)

....

- 127 R
- 400 R
- 200 R
- 560 R

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In the diagram shown, the difference in the two tubes of the manometer is 5 cm, the cross-section of the tube at A and B is $6\ \text{mm}^2$ and $10\ \text{mm}^2$ respectively. The rate at which water flows through the tube is $(g = 10\ \text{m s}^{-2})$

- $7.5\ \text{cc s}^{-1}$
- $12.5\ \text{cc s}^{-1}$
- $8.0\ \text{cc s}^{-1}$
- $10.0\ \text{cc s}^{-1}$

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Consider the following statements about a solenoid.

1. The magnetic field strength in a solenoid depends upon the number of turns per unit length in the solenoid.
2. The magnetic field strength in a solenoid depends upon the current flowing in the wire of the solenoid.
3. The magnetic field strength in a solenoid depends upon the diameter of the solenoid.

Which of the statements given above are correct?

- ○ 1, 2 and 3
- ○ 1 and 3
- ○ 2 and 3
- ○ 1 and 2

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A Zener diode voltage regulator operated in the range 120 V – 180 V produces a constant supply of 110 V and 250 mA to the load. If the maximum current is equally shared between the load and the Zener diode, then the values of load resistance (R_L) and series resistance (R_S) are respectively.

- $R_L = 280\Omega, R_S = 70\Omega$
- $R_L = 440\Omega, R_S = 140\Omega$
- $R_L = 70\Omega, R_S = 280\Omega$
- $R_L = 440\Omega, R_S = 1400\Omega$

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An electric dipole consists of two opposite charges of magnitude $q = 1 \times 10^{-6}$ C separated by 2 cm. The dipole is placed in an external field of 1×10^{5} N C^{-1}. What maximum torque does the field exert on the dipole? How much work must an external agent do to turn the dipole end for end, starting from position of alignment ($\theta = 0^{0}$)

- 4.4×10^{6} N m, 3.2×10^{-4} J
- -2×10^{3} N m, -4×10^{3} J
- 4×10^{3} N m, 2×10^{-3} J
- 2×10^{-3} N m, 4×10^{-3} J

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Two objects are initially at rest on a frictionless surface. Object 1 has a greater mass than object 2. The same constant force starts to act on each object. The force is removed from each object after it accelerates over a distance d. After the force is removed from both objects, which statement is correct (p: momentum, K: kinetic energy)?

- ○ $p_1 < p_2$
- ○ $p_1 > p_2$
- ○ $K_1 > K_2$
- ○ $K_1 < K_2$

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If the magnetic monopole exists, then which of the Maxwell's equation to be modified?

- $\oint \vec{E}.\overrightarrow{dA} = \frac{Q_{enclosed}}{\varepsilon_0}$
- $\oint \vec{B}.\overrightarrow{dA} = 0$
- $\oint \overrightarrow{B}.\overrightarrow{dl} = \mu_0 I_{enclosed} + \mu_0 \varepsilon_0 \frac{d}{dt} \int_s \vec{E}.\overrightarrow{dA}$
- $\oint \vec{E}.\overrightarrow{dl} = -\frac{d}{dt}\phi_B$

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A ball of mass m is moving towards a batsman at a speed v. The batsman strikes the ball and deflects it by an angle θ without changing its speed. The impulse imparted to the ball is given by-

- $mv \cos \theta$
- $mv \sin \theta$
- $2mv \cos(\theta/2)$
- $2mv \sin(\theta/2)$

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A coil having inductance L and resistance R is connected to a battery of emf ε at $t = 0$. If t_1 and t_2 are the time for 90% and 99% completion of the current growth in the circuit, then $10 \times \frac{t_1}{t_2}$ will be -

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Position of a $2\ \text{kg}$ mass moving along the X-axis is given by $x = 2\cos[2t]\text{m}$. Then maximum kinetic energy of the mass in joule is-

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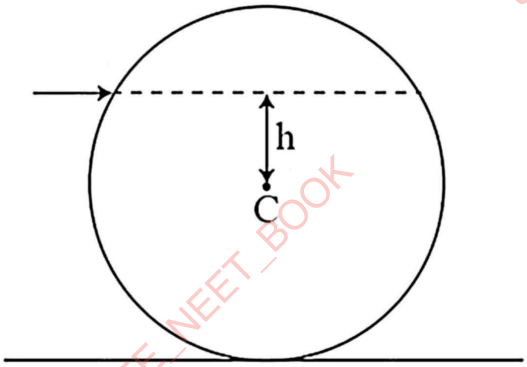
A galvanometer of resistance G is converted into an ammeter using a shunt of resistance R. If the ratio of the heat dissipated through the galvanometer and shunt is $3:4$, then R equals to $KG/8$. Find integral value of K.

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A point charge $+q$ is fixed at point A on the circumference of a fixed smooth non-conducting circular wire of radius R in a gravity free space. A small ring of mass m and charge $+q$ which can slide freely on the wire track is given tangential velocity at point B which is diametrically opposite to A. If the distance of closest approach between the fixed charge and the ring is R, then force applied by the wire track on the ring just after the release will be xF, find value of $12x$.
$\left[F = \frac{1}{4\pi\epsilon_0}\frac{q^2}{R^2}\right]$.

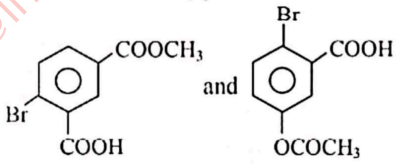
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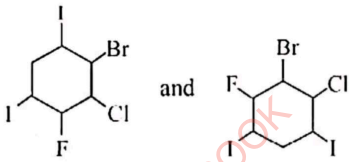
A billiard ball has mass $M = 250$ g and radius $R = 2.5$ cm and is initially at rest. A rod held horizontal at a height h above centre C hits the ball. The ball begins to roll without slipping. Find the value of h (mark answer in cm)

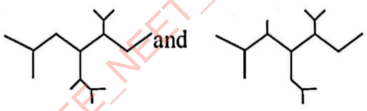


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Which of the following pair/s have same IUPAC naming :

○ 

○ 

○ 

○ None of these

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Match the following :

	List-I		List-II
(A)	Oxyhemoglobin	(i)	Hardness of water
(B)	EDTA	(ii)	Oxygen carrier
(C)	Hemoglobin	(iii)	Photosynthesis
(D)	Chlorophyll	(iv)	Oil of winter green
		(v)	Fe^{2+} paramagnetic

The correct match is

A B C D

- (v) (i) (ii) (iii)
- (iv) (ii) (i) (iii)
- (iii) (i) (ii) (iv)
- (v) (ii) (iii) (i)

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When N_2 is converted into N_2^+ and O_2 is converted into O_2^+, then the bond strength, respectively.

- ○ increases, increases
- ○ decreases, decreases
- ○ increases, decreases
- ○ decreases, increases

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Which of the following statement is/are true -

(I) Borazine is aromatic

(II) There are four isotopic disubstituted borazine molecules $B_3N_3H_4X_2$

(III) Borazine is more reactive towards addition reactions than benzene

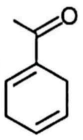
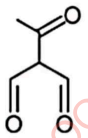
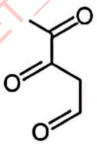
(IV) Banana bonds in B_2H_6 are longer than normal $B-H$ bonds

- ○ I, II and III
- ○ I, II and IV
- ○ I , II , III and IV
- ○ Only II

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Which of the following option(s) is/are correct for the products of the given reaction sequence?

Acetophenone $\xrightarrow{\text{Na in liquid } NH_3}$ H $\xrightarrow{O_3}$ I $\xrightarrow{Zn/H_2O}$ J (contains 5C atoms) + K

- Compound H is 
- Compound J is 
- Compound J is 
- None of these

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Ferrous ion charges to X ion, on reacting with acidified hydrogen peroxide.

The number of d-electrons present in X and its magnetic moment (in BM) are, respectively :

- ○ 6 and 6.93
- ○ 5 and 5.92
- ○ 5 and 4.9
- ○ 4 and 5.92

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Given below are two statements

Assertion (A): Hydrolysis of DNA does not form equal number of A and T; G and C

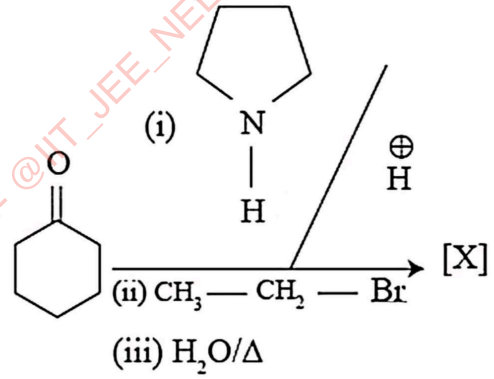
Reason (R): In DNA adenine forms hydrogen bonds with thymine and cytosine forms hydrogen bonds with guanine

The correct answer is-

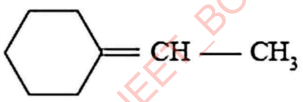
- Both A and R are correct and R is the correct explanation of A
- Both A and R are correct but R is not the correct explanation of $\dot{A}$
- A is correct but R is incorrect
- A is incorrect but R is correct

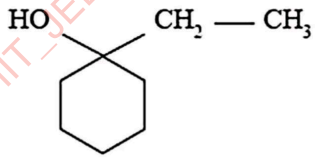
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In the reaction sequence



[X will be]

○ 

○ 

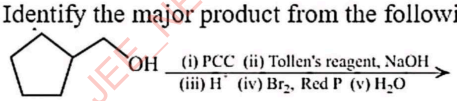
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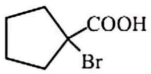
Rusting of iron is catalysed by which of the following?

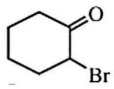
- ○ Fe
- ○ Zn
- ○ O_2
- ○ H^+

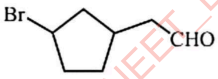
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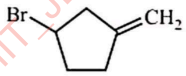
Identify the major product from the following reaction sequence



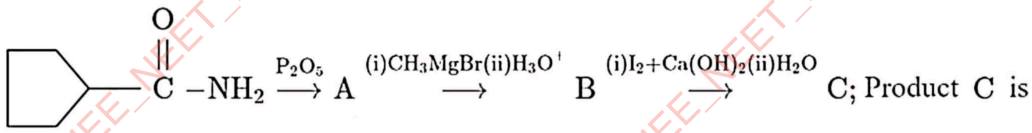
○ 

○ 

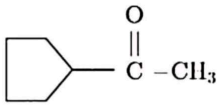
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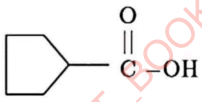
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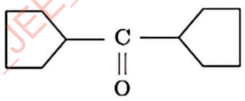
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C; Product C is

○ 

○ 

○ 

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In an atom, two electrons revolve around the nucleus in circular orbits of radii r and $4r$. The ratio of the time taken by them $(T_2 : T_1)$ to complete one revolution is....

- ○ 1 : 4
- ○ 4 : 1
- ○ 1 : 8
- ○ 8 : 1

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Which of the following is a correct statement?

- Aqueous solutions of Cu^{+} and Zn^{2+} are colourless
- Aqueous solutions of Cu^{2+} and Zn^{2+} are colourless
- Aqueous solution of Fe^{3+} is green in colour
- Aqueous solution of MnO_4^- is colourless

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Consider the following reactions, which are carried out at the same temperature.

$$CH_3 - Br + OH^{\ominus} \xrightarrow{EtOH} CH_3 - OH + Br^{\ominus} \quad (i)$$

$$CH_3 - Br + OH^{\ominus} \xrightarrow{DMSO} CH_3 - OH + Br^{\ominus} \quad (ii)$$

Which of the following statement is correct about these reactions?

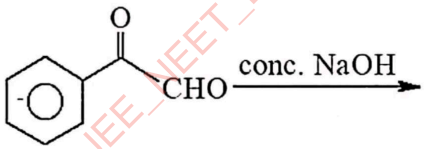
- Both the reactions take place at the same rate
- The second reaction takes place faster than first reaction
- The first reaction takes place faster than second reaction
- Both the reactions take place by S_N1 mechanism.

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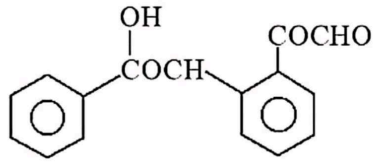
Which of the following statements are correct ?
1) The elements like F, Cl, O etc having high values of electron affinity act as strong oxidizing agents.
2) The elements having low values of ionisation energies act as strong reducing agent.
3) The formation of S^{2-} is an endothermic process.
4) Formation of Cl^{-} is exothermic whereas formation of O^{2-} is endothermic.

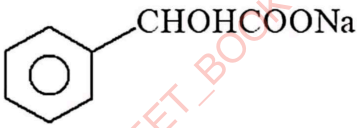
- (1), (2) and (3)
- (1) and (3)
- (1), (2), (3) and (4)
- (1), (2) and (4)

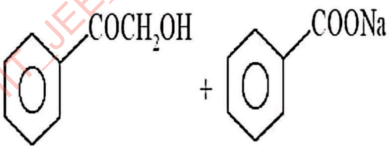
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Product should be

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Which among the given statements are true?
(i) PH_5 and $BiCl_5$ does not exist
(ii) $p\pi - d\pi$ bond is present in SO_2
(iii) I_3^+ has bent geometry according to VSEPR Theory
(iv) SeF_4 and CH_4 have same shape and geometry

- i, ii, iii
- i, iii
- i, iii, iv
- i, ii, iv

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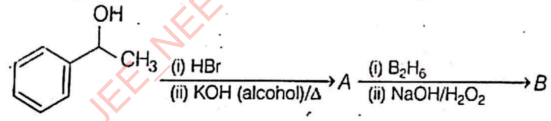
The correct order for increasing reactivity towards S_N1 substitution for the following compounds is

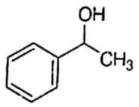
1. CH_3CH_2Br 2. $CH_2 = CH - CH(Br)CH_3$
3. $CH_2 = CHBr$ 4. $CH_3 - CH(Br)CH_3$

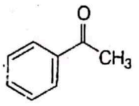
- $1 < 4 < 3 < 2$
- $4 < 1 < 2 < 3$
- $3 < 1 < 2 < 4$
- $3 < 1 < 4 < 2$

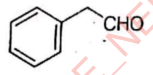
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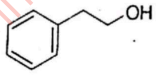
The major product 'B' of the following reaction sequence is



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Sodium fusion extract of aniline when heated with ferrous sulphate solution and then acidified with concentrated H_2SO_4 from which of the following complexes?

- $[Fe(CN)_6]^{4-}$
- $Fe_4[Fe(CN)_6]_3 \cdot xH_2O$
- $[Fe(SCN)]^{2+}$
- $[Fe(CN)_5NOS]^{4-}$

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$\frac{x}{20}$ M concentration of H^+ion must be maintained in a saturate $H_2S(0.1M)$ to precipitate CdS but not ZnS, if $\left[\mathrm{Cd}^{2+}\right] = \left[\mathrm{Zn}^{2+}\right] = 0.1\mathrm{M}$ initially.
$\mathrm{K_{sp}(CdS)} = 8 \times 10^{-27}$, $\mathrm{K_{sp}(ZnS)} = 1 \times 10^{-21}$ $\mathrm{K_a(H_2\ S)} = 1 \times 10^{-21}$. ZnS will not precipitate at a concentration of H^+greater than $\frac{\mathrm{x}}{20}$ M. The value of x is.

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Calculate the difference in the heat of formation (in calories) of carbon monoxide at constant pressure and at constant volume at $27^{\circ}C$.

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A metal plate of dimension of $(1 \times 2\ \text{cm}^2)$ has to be coated on both the sides by Cu metal. How long does it take to deposit Cu of 0.01 cm thickness, if 1.5 A current is used? [Electrochemical equivalence of Cu is $0.0003\ \text{g/C}$ and the density of Cu is $9\ \text{g/cm}^3$] Mark answer in seconds

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How many isomeric alcohols of molecular formula $C_5H_{12}O$ give blue colour in victor - Maeyer test finally?

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An acid ' A ' is 0.4 N. In a neutralization process 50ml of ' A ' is mixed with 30ml of sodium carbonate solution (42.4 g/Lt) and 10 mL of the resulting solution is diluted to 50 mL and then 5 mL of this solution (B) is completely neutralised with certain volume of 40 times diluted solution of (A). find the volume of diluted solution of (A) required for the neutralization of B (in mL)

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Let A be a square matrix of order two such that $A - AA^T = I_2$. If trace $(A) \neq 1$ and entries are complex numbers, then number of such matrices are
(where trace(A) denotes sum of elements of principal diagonal of matrix A)

- 4
- 5
- 2
- 3

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If $f(x) = \lim_{n\to\infty} n^2\left(x^{\frac{1}{n}} - x^{\frac{1}{n+1}}\right), x > 0$, then $\int xf(x)dx =$

- $\frac{x^2}{2}\log x + C$
- $\frac{x^2}{2}\log x + \frac{x^2}{4} + C$
- $\frac{x^2}{2}\log x - \frac{x^2}{4} + C$
- $-\frac{x^2}{2}\log x + \frac{x^2}{4} + C$

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The fractional part of $\frac{2^{78}}{31}$ is____

- $\frac{2}{31}$
- $\frac{4}{31}$
- $\frac{6}{31}$
- $\frac{8}{31}$

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If the equation $x^4 + 4x^3 - 2x^2 - 12x + k = 0$ has 4 real roots, then

- $-1 < k < 10$
- $-7 < k < 7$
- $-7 < k < 9$
- $1 < k < 9$

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Let $\alpha, \beta \in R$, satisfy the equations $\cot^2\alpha - \text{cosec}^2\beta = \frac{2k}{3} - 5$ & $\cot^2\beta - \text{cosec}^2\alpha = \frac{k^2}{2}$, then product of all possible value(s) of k is equal to

- $-\frac{4}{3}$
- -6
- -1
- $\frac{2}{3}$

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Let A and B be two 3×3 invertible matrices such that A is an idempotent matrix and $\det(\text{adj}\, B) = \det(A)^{12} + \det\left(A - A^{\top}\right)^{13} + \det(\text{adj}\, A - I)^{14} + L$

where $L = \lim_{x \to 0} \frac{\int_0^{\sin^2 x} \ln(1+t^3)\,dt}{\int_0^x (e^t - 1 - t)(\tan t - t)(1 - \cos t)\,dt}$

Then the value of $\det(B)$

- 4
- 3
- 2
- 5

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If $f(x) = \frac{ax+b}{bx-a}, x \neq \frac{a}{b}, a > 0, b > 0$ and if $g(x) = f(f(x)) + f\left(f\left(\frac{1}{x}\right)\right)$ and $g : [1, \infty) \to [2, \infty)$ then the value of x for which $g(x) = \frac{5}{2}$, is

- 4
- 3.5
- 2
- 4.5

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Let a, b and c be non-negative real numbers satisfying $a + b + c = 9$. If the maximum value of the expression $a^2b^3c^4$ can be expressed as 2^x3^y, where x and y are natural numbers, then the value of $\log_{10}(x^y)$, is:

- 2
- 3
- 4
- 6

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Let S be the relation on N defined by $S = \left\{(x, y) : \frac{y}{2} \leq x \leq 2y\right\}$. Then, S is

- reflexive only
- transitive only
- symmetric only
- reflexive and symmetric

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Let $A,\ B$ and C be three points taken on the parabola $y^2 = 4ax$ with coordinates $\left(at_i^2, 2at_i\right)\ i = 1,\ 2,\ 3$, where $t_1,\ t_2$ and t_3 are in AP. If $AA',\ BB'$ and CC', are focal chords and coordinates of A', B', C' are $\left(at_i'^2, 2at_i'\right),\ i = 1,\ 2,\ 3$ then $t_1',\ t_2'$ and t_3' are in

- AP
- GP
- HP
- AGP

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If $\vec{a} = 2\hat{i} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$, $\vec{c} = 4\hat{i} - 3\hat{j} + 7\hat{k}$, then the vector $\vec{r}$ satisfying $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$ is

- $\hat{i} + 8\hat{j} + 2\hat{k}$
- $\hat{i} - 8\hat{j} + 2\hat{k}$
- $\hat{i} - 8\hat{j} - 2\hat{k}$
- $-\hat{i} - 8\hat{j} + 2\hat{k}$

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A purse contains 10 ten rupee coins and 5 five rupee coins. Two coins are randomly drawn. If the expected value of 2 drawn coins is λ, then $\frac{9\lambda}{4}$ is equal to

- 34.5
- 37.5
- 39.5
- 33.5

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Area bounded by $f(x) = \min(\{x-2\}, \{3-x\}), x = 4, x = -4$ and x-axis is equal to (where $\{.\}$ denotes fractional part)

- 2.5
- 2
- 1.5
- 1

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Let $f(x)$ be a polynomial satisfying $f(x)f\left(\frac{1}{x}\right)+5-3f(x)-3f\left(\frac{1}{x}\right)=0, \forall x \in R-\{0\}$ and $f(2)=11$, then $f(3)$ is equal to:

- 21
- 12
- 20
- 11

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If the chord of the circle $x^2 + y^2 - 4x - 2y - c = 0$ is trisected at the points $\left(\frac{1}{3}, \frac{1}{3}\right)$ and $\left(\frac{8}{3}, \frac{8}{3}\right)$ then :

- Length of the chord $= 7$
- $c = 20$
- Radius of circle $= 6$
- $c = 25$

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Consider the system of equations: $x\sin\theta - 2y\cos\theta - az = 0$, $x + 2y + z = 0$ and $-x + y + z = 0,\ \theta \in R$.

- The given system will have infinite solutions for $a = 2$.
- The number of integer values of a is 3 for the system to have non-trivial solutions.
- For $a = 1$, there exists θ for which the system will have unique solution.
- For $a = 3$, there exists θ for which the system will have a infinite solutions.

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If $A\left(\vec{a}\right), B\left(\vec{b}\right), C\left(\vec{c}\right)$ and $D\left(\vec{d}\right)$ are four points such that $\vec{a} = -2\hat{i} + 4\hat{j} + 3\hat{k}, \vec{b} = 2\hat{i} - 8\hat{j}, \vec{c} = \hat{i} - 3\hat{j} + 5\hat{k}$ and $\vec{d} = 4\hat{i} + \hat{j} - 7\hat{k}$ and d_1 is the shortest distance between the lines AB and CD, then which of the following is true:

- ○ $d_1 = 0$, hence AB and CD intersect
- ○ AB and CD are skew lines and $d_1 = \frac{20}{13}$
- ○ AB and CD are skew-lines and $d_1 = \frac{23}{13}$
- ○ AB and CD are skew lines and $d_1 = \frac{25}{13}$

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Let $I_n = \int_{-1}^{1} |x| \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \ldots + \frac{x^{2n}}{2n}\right) dx$ where $n \in N$. If $\lim_{n\to\infty} I_n$ can be expressed as a rational number $\frac{p}{q}$ in the lowest form, then find the value of $p + q$.

- 3
- 7
- 5
- 8

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Find sum of all possible values of the real parameter ' b ' if the difference between the largest and smallest values of the function $f(x)=x^2-2bx+1$ in the segment $[0,1]$ is 4 .

- 5
- 1
- 3
- 2

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Let $y = f(x)$ satisfies the differential equation, $xy(1+y)dx = dy$. If $f(0) = 1$ and $kf(2) = (1+f(2))e^2$, where $k \in N$, then k is equal to

[Note : e denotes Napier's constant]

- 1
- 3
- 2
- 4

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n is selected from the set $\{1, 2, 3, \ldots\ldots\ldots 100\}$ and the number $2^n + 3^n + 5^n$ is formed. Let the total number of ways of selecting n is P, so that the formed number is divisible by 4, then $\frac{P}{7}$.

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Let $2\vec{a} = \vec{b} \times \vec{c} + 2\vec{b}$ where $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three unit vectors, then sum of all possible values of $\left|3\vec{a} + 4\vec{b} + 5\vec{c}\right|$ is

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A complex number z satisfies $arg\left(\frac{z}{z-i}\right)=\frac{\pi}{3}$ and $|z|=|z-i|$, then evaluate $[Re\,(2z-i)]$ where $[\cdot]$ represents the greatest integer function.

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If $\left(1+x^2\right)^2(1+x)^n=\sum_{k=0}^{n+4} a_k x^k$ where a_1, a_2, a_3 are in A.P., then sum of possible values of n is

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The distance (in units) between the circumcentre and the centroid of the triangle formed by the vertices $(1, 2)$, $(3, -1)$ and $(4, 0)$, is $K\sqrt{2}$. Find 30K.

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